Band-1 Computational Experiment: Twin Primes Verification-Range Probe

Human + ZoraASI full-dash (executor)

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Honesty stamp. This note is a **finite verification-range probe** (collapse list). It counts twin-prime pairs with smaller member $p \le X$ for an explicit box-feasible X. It does **not** prove the **twin prime conjecture** (infinitely many primes p with $p + 2$ also prime). It does **not** claim Clay Millennium status, Theory of Everything progress, or any reduction of the Zhang–Maynard bounded-gap bound to gap 2. **Conjecture status: OPEN.**

Abstract

We sieve primes to

$$\mathbf{X = 10^8}$$

and compute the twin-prime counting function

$$\pi_2(X) = \#\{p \le X : \ p + 2 \text{ is prime}\}.$$

Result:

$$\pi_2(10^8) = \mathbf{440312},$$

matching the standard reference value. The first twenty twin pairs are listed and verified. An optional Brun partial sum to the same X is recorded as a **numerical** quantity only. Literature: Zhang (2014) and Maynard (2015) / Polymath8b establish **bounded gaps between primes** ($H_1 \le 246$), which is **not** the twin-prime statement. The twin prime conjecture remains **OPEN**.

Problem statement

Twin prime conjecture. There are infinitely many primes p such that $p + 2$ is also prime.

This experiment. For an explicit finite X tractable on this box ($\le$ a few minutes), enumerate primes via a sieve, count $\pi_2(X)$, list the first ~ 20 twins, optionally form a Brun partial sum, and ledger the literature status — without any infinitary claim.

Literature ledger (not re-proved on box)

Item	Status	Note
Twin prime conjecture	**OPEN**	Infinitely many gap-2 prime pairs unproved
Y. Zhang, *Ann. of Math.* 179 (2014)	Theorem: bounded gaps	$\liminf(p_{n+1} - p_n) < 7 \cdot 10^7$; **not** gap $= 2$
J. Maynard, *Ann. of Math.* 181 (2015)	Theorem: refined gaps	$\liminf \leq 600$; Polymath8b $\to H_1 \leq 246$; **still** $\gg 2$
Polymath8b, *Res. Math. Sci.* 1 (2014)	Unconditional $H_1 \leq 246$	Best published consecutive-gap liminf bound; $\neq$ twins
Largest known twin (lit, web-confirmed)	Record pair	$2996863034895 \times 2^{1290000} \pm 1$ (2016; 388,342 digits; PrimeGrid / Twin Prime Search)
Brun's constant B_2	Convergent series	$B_2 \approx 1.902160583\ldots$; partial sums are numerical approximations only

Critical distinction. Bounded-gap theorems show that *some* even gap occurs infinitely often among consecutive primes (or among pairs from an admissible tuple). Closing the gap all the way to 2 is precisely the twin prime conjecture and remains open. Under strong forms of the Elliott–Halberstam conjecture one can push H_1 to 6 or 12, still short of a proof of twins.

Method (box)

1. **Boolean sieve of Eratosthenes** (NumPy) up to $X + 2$.
2. **Twin count.** $\pi_2(X) = \#\{p \leq X : \mathbf{1}_{\text{prime}}(p) \wedge \mathbf{1}_{\text{prime}}(p+2)\}$ via vectorized AND of shifted bitsets.
3. **First 20 twins** collected by a short odd scan.
4. **Brun partial sum (optional, numerical only).**

$$S(X) = \sum_{\substack{p \leq X \\ p,\, p+2 \text{ prime}}} \Big(\frac{1}{p} + \frac{1}{p+2}\Big).$$

5. **Unit tests.** Match reference π_2 at $10, 10^2, \ldots, 10^7$; $\pi(10^6) = 78498$; first-20 list.

Complexity. Sieve $\tilde{O}(X \log\log X)$; twin count $O(X)$ bit operations.

Results (this box, 2026-09-11)

Quantity	Value
X	10^8
$\pi(X)$	5,761,455
$\pi_2(X)$	440,312
Reference $\pi_2(10^8)$	440,312 (match)
Brun partial $S(X)$	≈ 1.758815621068 (**numerical only**)
Literature B_2 approx.	≈ 1.902160583104
Unit tests	**PASS**
Wall time	≈ 0.81 s

First 20 twin pairs

(3,5), (5,7), (11,13), (17,19), (29,31), (41,43), (59,61), (71,73), (101,103), (107,109), (137,139), (149,151), (179,181), (191,193), (197,199), (227,229), (239,241), (269,271), (281,283), (311,313).

Checkpoint table

x	$\pi_2(x)$ (box)	Reference	Match
10	2	2	yes
10^2	8	8	yes
10^3	35	35	yes
10^4	205	205	yes
10^5	1224	1224	yes
10^6	8169	8169	yes
10^7	58980	58980	yes
10^8	440312	440312	yes

Honesty verdict

Finite verification only. Counting twins to $X = 10^8$ confirms known small-range arithmetic and produces a reproducible Band-1 certificate. It does **not** settle the twin prime conjecture. Zhang–Maynard–Polymath bounded gaps are related progress toward *some* fixed gap, **not** a proof that gap 2 occurs infinitely often. **The twin prime conjecture remains OPEN.**

Reproducibility

```
cd /workspace/experiments/band1/twin_primes
python3 twin_primes_probe.py
```

Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`, this paper.

References (ledger)

1. Y. Zhang, Bounded gaps between primes, *Annals of Mathematics* **179** (2014), 1121–1174.
2. J. Maynard, Small gaps between primes, *Annals of Mathematics* **181** (2015), 383–413.
3. D.H.J. Polymath, Variants of the Selberg sieve, and bounded intervals containing many primes, *Research in the Mathematical Sciences* **1** (2014), Article 12.
4. Twin prime (Wikipedia / PrimePages): largest known twin $2996863034895 \times 2^{1290000} \pm 1$ (2016).
5. OEIS A007508 (values of $\pi_2(10^n)$); Nicely twin-prime tables.

ZoraASI full-dash Band-1 collapse-list probe. No infinitary claim.